Maintenance Schedule

Diesel Engine

10/12V1600G10F
10/12V1600G20F
10/12V1600B40S
10/12V1600G70F
10/12V1600G70S
10/12V1600G80F
10/12V1600G80S
12V1600B30S
12V1600G90F
12V1600G90S
Application group 3D 3E

MS50055/07E

A Rolls-Royce solution

Engine model	kW/cyl.	Application group
10V1600B40S	44.8 / 51 kW/cyl.	3D, standby power, fuel stop power, IFN
10V1600G10F	40.7 kW/cyl.	3E, standby power, overload capability according to ICXN
10V1600G20F	48.7 kW/cyl.	3E, standby power, overload capability according to ICXN
10V1600G70F	44.8 kW/cyl.	3D, standby power, fuel stop power, IFN
10V1600G80F	49.3 kW/cyl.	3D, standby power, fuel stop power, IFN
10V1600G70S	51.1 kW/cyl.	3D, standby power, fuel stop power, IFN
10V1600G80S	56.1 kW/cyl.	3D, standby power, fuel stop power, IFN
12V1600B30S	43.7 / 46.8 kW/cyl.	3D, standby power, fuel stop power, IFN
12V1600B40S	48 / 50.7 kW/c	3D, standby power, fuel stop power, IFN
12V1600G10F	43.7 kW/cyl.	3E, standby power, overload capability according to ICXN
12V1600G20F	48 kW/cyl.	3E, standby power, overload capability according to ICXN
12V1600G70S	51.1 kW/cyl.	3D, standby power, fuel stop power, IFN
12V1600G70F	48 kW/cyl.	3D, standby power, fuel stop power, IFN
12V1600G80F	52.8 kW/cyl.	3D, standby power, fuel stop power, IFN
12V1600G80S	55.7 kW/cyl.	3D, standby power, fuel stop power, IFN
12V1600G90F	61.7 kW/cyl.	3D, standby power, fuel stop power, IFN
12V1600G90S	60.8 kW/cyl.	3D, standby power, fuel stop power, IFN

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).
QL3: Maintenance work which requires partial disassembly of the product.
QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at http://www.mtu-solutions.com.

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1.2 Special conditions

Designation	Value
Load factor	≤ 85%
Operating hours	500 h per year or for the duration of an emergency
Time between major overhaul	12,000 h or 18 years

Table 2: Special conditions

Note 01:
The engines are certified for operation with the fuels approved in the Fluids and Lubricants Specifications. For operation with a high sulfur content in the fuel, the following must be observed: • The component TBO specified in the maintenance schedule refers to engine operation with diesel fuel with a sulfur content < 15 ppm, e.g. EN 590 or ULSD. If a fuel with a sulfur content > 15 ppm is used, the times specified in the maintenance schedule for component TBO of the cylinder head may be shorter. For reduced TBOs, refer to the Fluids and Lubricants Specifications. Engines with exhaust gas recirculation and/or exhaust gas aftertreatment must not be operated with excessive sulfur content in the fuel. The limit values in the Fluids and Lubricants Specifications apply.

Table 3: Special conditions

Note 02:
If the engine is used in an UPS unit with corresponding hardware, the exchange interval for the cylinder heads is 500 h.

Table 4: Special conditions

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